

Nadir Zouaoui

Electrical Engineer

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Profile

State engineer specialized in renewable energy, with four years of experience in Oil & Gas and solar. I design, wire and commission electrical systems: skids, gas-pipeline SCADA, ATEX cabinets and PV plants, with over 1 MWp of installed solar capacity delivered.

Professional experience

Electrical Engineer, RMBTECH EURL

Mar 2025 – Present

- **SCADA for Sonatrach gas pipelines.** Led the electrical engineering: site audit of 66 isolated stations (solar or grid powered) on GK3, which produced a detailed equipment condition status for each station (good, damaged, vandalized, etc.); deployment on GK1/GK2; pre-engineering of GR1/2/4/6 (integrator: SUPCON).
- **Critical equipment and ATEX.** Engineering for the replacement of an 8500 A rectifier for the AOA electrolysis system; design of an ATEX Ex "p" pressurized cabinet for Sonatrach GP2Z (Arzew).
- **Industrial rehabilitation.** Technical direction, redesign in Eplan, rewiring and automatic commissioning (Siemens S7-1200) of a BOSSAR packaging machine for GBFoods (Jumbo).
- **Agro-industry.** Electrical design for the SCADA system of the OAIC grain silo in Mesra.

Electrical Engineer, PROSKID EURL

Jul 2023 – Mar 2025

- Designed and sized the electrical systems of Oil & Gas skids for clients ENGCB/GCB and SARPI, and the off-grid solar power supplies feeding them.
- Produced power supply, power and control schematics in EPLAN Electric P8 and AutoCAD Electrical.
- Commissioned 27 Milton Roy chemical anticorrosion dosing skids and their solar installations.
- **ATEX Ex "d" flameproof enclosure.** Designed and built it for a corrosion inhibitor skid, including component mounting and wiring.

Photovoltaic Design Engineer, Sungy SPA

Oct 2022 – Jul 2023

- Over 300 kWp of installed solar capacity.
- 3D layout, sizing and simulation of PV systems; electrical sizing and diagrams for each installation.
- Commissioning and monitoring; energy sobriety plans; deployable mobile solar solutions.

Freelance

5.4 MW industrial solar installation, SARL Ibrahim & Fils (IFRI)

Akbou, Béjaïa

- Rooftop PV design for a zero-injection self-consumption plant across several production sites: JinkoSolar 635 Wp bifacial modules, Huawei SUN2000-150K inverters, 10 × 241 kWh battery storage.

- Layout, string and inverter zoning, and DC/AC cable sizing optimized for voltage drop and cost.
- Single-line diagram dossiers, load and power quality analysis.

Electrical training modules, ROAR Training Solutions (Australian RTO)

- Developed first-person 3D playable exercises for electrical competency training, including a 3D substation exercise and a Low Voltage Rescue and CPR simulation (hazard identification, PPE, breaker isolation, insulated-hook rescue, then CPR with AED, guided by a phase-gated procedure checklist).
- 3D assets and character animation in Blender; interaction logic and training scenarios in Godot 4.
- Packaged as SCORM 1.2 modules running in the browser (WebGL2) for delivery through the client's LMS.

Motion design and video

Sept 2018 – Present

Freelance motion design and video work. Portfolio on Behance, Freelancer, Fiverr.

Thesis project

5 kW autonomous photovoltaic microgrid

May – Jul 2021

- Sized and built a microgrid with a 3 kW PV array, an 800 Ah battery bank, a 50 kVA diesel generator and a three-input hybrid inverter (PV, battery, grid/genset), with electrical and synoptic diagrams.
- Installed the outdoor (PV array, supports) and indoor (inverter, batteries, generator, boxes, ATS) equipment, then commissioned and tested the system.

Education

State Engineer + Master II

2018 – 2021

École Nationale Polytechnique d'Oran – Maurice Audin (ENPO-MA). Valedictorian 2018–2021.

Preparatory cycle in sciences and technologies

2016 – 2018

ENPO-MA

Certifications

Low-voltage design fundamentals + Caneco BT introduction

Sept – Oct 2025

Lokelect Engineering, 40-hour online course. Certificate ID 2b3376a9aea18854

[Verify the certificate](#)

Earlier experience

Document Controller, CEVIELEC SPA: ALRAR gas complex re-instrumentation	2022
Internship, SKTM: 12 MW Dhaya PV plant	2021
Internship, SDO Sonelgaz: MV-LV distribution network	2019
Internship, ENIE Solar: PV panel production	2019
Internship, CITAL: Sidi Bel Abbès tramway traction	2018

Skills and tools

Tools	EPLAN Electric P8 & Pro Panel, Archelios PV, AutoCAD Electrical, Caneco BT, TIA Portal / Step 7, Siemens S7-1200, Microsoft Excel, Blender 3D, Adobe Suite
Domains	Oil & Gas, industrial equipment, hazardous areas (ATEX), MCC and switchgear, commissioning, PV system design and sizing, electrical diagrams and schematics, SCADA, industrial automation
Languages	Arabic (native), English (fluent), French (bilingual, DALF C1, 76/100)
Other	Driving license B, national service exempt